

Question 1 (12 Marks) Begin a SEPARATE sheet of paper

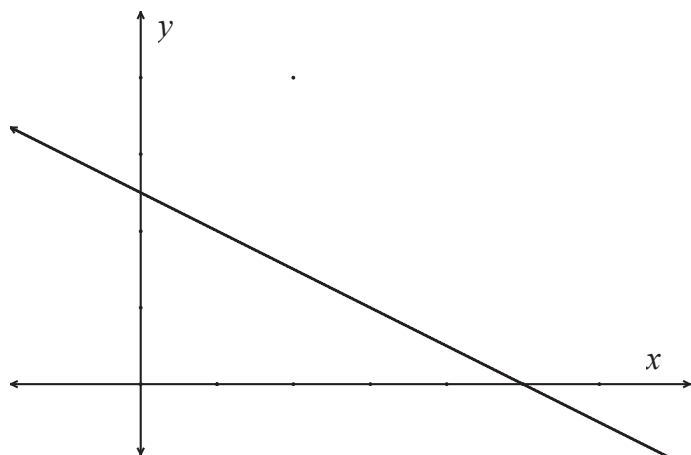
	Marks
(a) Evaluate $\frac{e+1}{\pi}$, correct to three decimal places.	2
(b) Find θ , to the nearest degree, if $\sin \theta = \frac{4 \sin 57^\circ}{6.7}$	2
(c) What is the centre and radius of a circle with equation $(x+2)^2 + (y-3)^2 = 2.25$.	2
(d) The mean of 3, 5, 7, x is 6.75. What is the value of x ?	1
(e) If $x = 2.35$, evaluate the expression $ -3 - 4x $	1
(f) Factorise $3x^2 - 5x - 2$.	1
(g) Express 2.32 radians as an angle in degrees, correct to the nearest minute.	1
(h) Write down the domain and range for $y = \sqrt{x^2 - 9}$.	2

Question 2 (12 Marks) Begin a SEPARATE sheet of paper

	Marks
(a) Differentiate the following:	
(i) $e^{0.5}$.	1
(ii) $\log_e (x^2 - 3x)$.	1
(iii) $\frac{2x-1}{\cos x}$	1
(b) Integrate: $\int \sin \frac{x}{2} dx$	2
(c) Evaluate: $\int_0^1 \frac{5}{6} e^{3x} dx$ (Leave your answer in exact form).	2
(d) Given $\log_m p = 1.75$ and $\log_m q = 2.25$. Find the following:	
(i) $\log_m pq$	1
(ii) $\log_m \frac{q}{p}$	1
(iii) $\sqrt[5]{pq^2}$ in terms of m .	3

Question 3 (12 Marks) Begin a SEPARATE sheet of paper**Marks**

The number plane shows the line n with equation $x + 2y = 5$.
The point P is $(2, 4)$ and O is the origin.



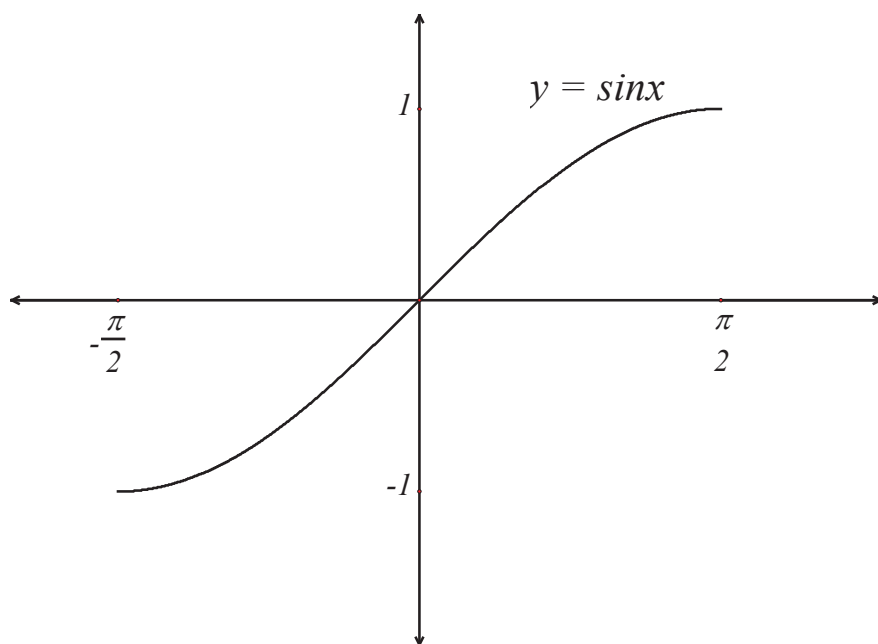
- | | |
|--|---|
| (a) Find the coordinates of the midpoint M , of the interval OP . | 1 |
| (b) Show that the point M lies on the line n . | 2 |
| (c) Find the gradient of the line OP . | 1 |
| (d) Show that the line n is the perpendicular bisector of the interval OP . | 3 |
| (e) Line n meets the x -axis at Q . Find the co-ordinates of Q . | 1 |
| (f) A line is drawn through O parallel to PQ and it meets line n in R . Find the co-ordinates of R . | 2 |
| (g) What sort of quadrilateral is $PQOR$? Give reasons for your answer. | 2 |

Question 4 (12 Marks) Begin a SEPARATE sheet of paper**Marks**

- (a) Let m and n be positive whole numbers where $m > n$.
- (i) Show that a triangle with sides $m^2 + n^2$, $m^2 - n^2$, $2mn$ obeys Pythagoras' Theorem. **2**
- (ii) Which Pythagorean Triad is generated when $m = 3$ and $n = 2$? **1**
- (b) Consider the function $f(x) = x - 2 \log_e x$, for $x > 0$.
- (i) Find the first **and** second derivative of $f(x)$. **2**
- (ii) Find the co-ordinates of the turning point(s) and determine their nature. **2**
- (iii) Show that there are no points of inflexion. **2**
- (iv) What is the maximum value of $x - 2 \log_e x$ in the domain $1 \leq x \leq 5$? **1**
- (v) Sketch the curve $y = x - 2 \log_e x$, for $0 < x \leq 5$. **2**

Question 5 (12 Marks) Begin a SEPARATE sheet of paper**Marks**

(a)



The diagram above shows the graph of $y = \sin x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.

(i) Copy the diagram onto your exam paper.

2

On the same set of axes, graph $y = \cos 2x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.

(ii) Show that the two graphs intersect where $x = \frac{\pi}{6}$.

1

(iii) Calculate the exact area enclosed between the two curves.

3

(b) Show that $\frac{\sec \theta - \sec \theta \cos^4 \theta}{1 + \cos^2 \theta} = \sin \theta \tan \theta$.

3

(c) (i) Find the value(s) of k for which $x^2 + (2 - k)x + 2.25 = 0$ has equal roots.

2

(ii) Find the value(s) of k for which $y = kx + 1$ is a tangent to $y = x^2 + 2x + 3.25$.

1

Question 6 (12 Marks) Begin a SEPARATE sheet of paper**Marks**

- (a) Boat A is 15km from point P on a bearing of 055° T.
Boat B is 25km from point P on a bearing of 135° T.
- (i) Draw a diagram showing the information above and find the angle APB . 2
- (ii) Calculate, to one decimal place, the distance that the two boats are apart. 2
- (b) Evaluate $\sum_{r=3}^7 2^r - 3r$. 2
- (c) Water enters an empty container where the rate of filling is $R = 6(e^t + e^{-t})$ where $R = \frac{dV}{dt}$ in litres per minute. The volume (V) of water in the vessel is in litres and time (t) is in minutes.
- (i) What is the initial rate of filling? 1
- (ii) Find V in terms of t . 2
- (iii) Show that, when the container holds 5 litres, then $6e^{2t} - 5e^t - 6 = 0$. 1
- (iv) Hence, find to the nearest second, the time taken for the volume to reach 5 litres. 2

Question 7 (12 Marks) Begin a SEPARATE sheet of paper**Marks**

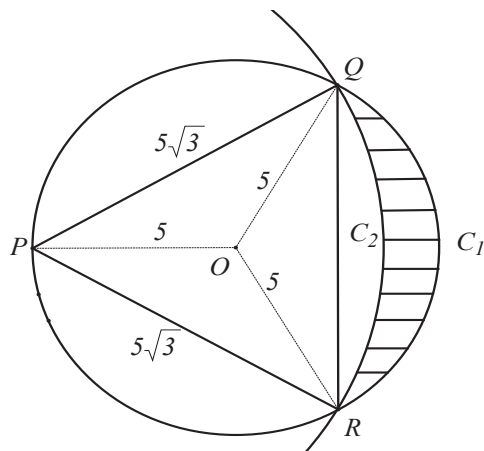
(a) Find y if $\frac{dy}{dx} = \frac{3\sec^2 x}{\tan x}$ and $y = 4$ when $x = \frac{\pi}{4}$. 3

(b) *Twinkle Finance* offers its investors the opportunity to have interest credited to their investment “as often as you wish”. Naturally, many investors decide for the “EVERY MINUTE” plan in which *Twinkle* offers 12% pa, compounded each minute.

(i) Tanya invests \$1000 for a year with *Twinkle* on the “EVERY MINUTE” plan. Theoretically, *Twinkle*’s computers multiply Tanya’s balance by approximately 1·000 000 228 every minute. Show why this is so. 2

(ii) How much, to the nearest dollar, is Tanya’s investment worth after 1 year? 1

(c)



In the diagram above, an equilateral triangle PQR is inscribed in a circle C_1 , with centre O , and radius 5 units. $\angle POQ = \angle QOR = \angle POR$.

An arc of another circle C_2 , with centre P , and radius $5\sqrt{3}$ units, intersects C_1 at the points Q and R , as shown in the diagram.

(i) Show $\angle QPO = \frac{\pi}{6}$, giving reasons. 1

(ii) Find the area of the sector QPR , of the circle C_2 . 1

(iii) Find the area of the sector QOR , of the circle C_1 . 1

(iv) Hence, or otherwise, show that the area of the shaded region is $25\left(\frac{\sqrt{3}}{2} - \frac{\pi}{6}\right)$ units². 3

Question 8 (12 Marks) Begin a SEPARATE sheet of paper**Marks**

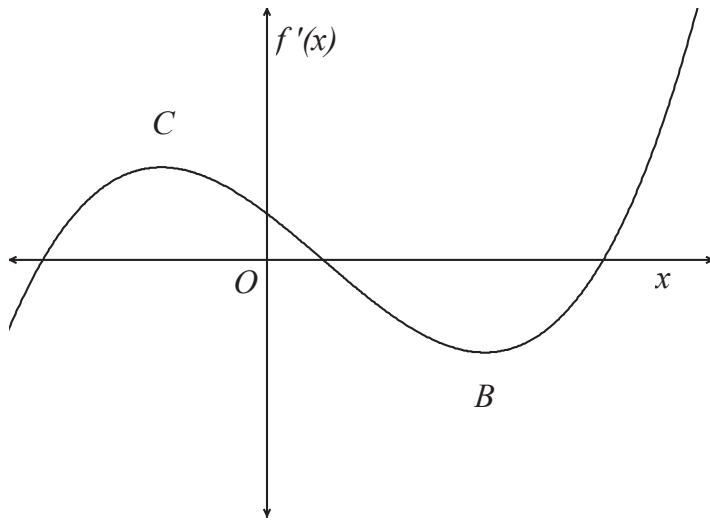
- (a) Let A be the point $(-2, 0)$ and B be the point $(6, 0)$.
At $P(x, y)$, PA meets PB at right angles.
- (i) Show that the gradient of PA is $\frac{y}{x+2}$. 1
- (ii) Find the equation of the locus of P **and** give a geometric description of the locus. 3
- (b) The velocity of an object is given by the equation $v = 6t - 8 - t^2$, where time (t) is in seconds and velocity (v) is in metres/second. Initially, the object is 5 metres to the right of the origin.
- (i) Find an equation for the displacement of the object. 2
- (ii) Find when the object is momentarily at rest. 1
- (iii) Find the distance travelled by the object in the first four seconds. 2
- (c) Two dice are biased so that, the probability of throwing a six face up is $\frac{3}{8}$ and the probability of throwing any other number is $\frac{1}{8}$.
- Find the probability of:
- (i) Rolling a double six face up. 1
- (ii) Rolling the dice so that neither is a six face up. 1
- (iii) Only one six appearing face up when the two dice are rolled. 1

Question 9 (12 Marks) Begin a SEPARATE sheet of paper**Marks**(a) (i) Write $x^2 - 6x + 8 = 2y$ in the form $(x - h)^2 = 4a(y - k)$.**2**

(ii) Hence, state the co-ordinates of focus for the parabola.

1

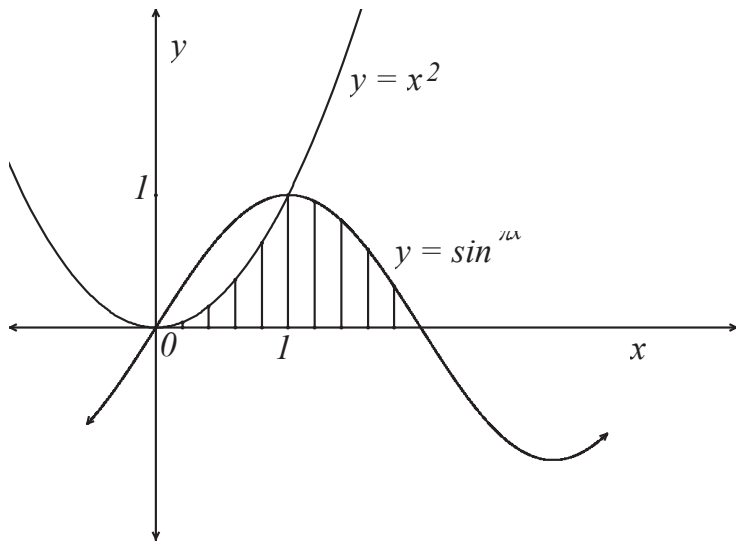
(b)



The graph of $y = f'(x)$ is shown above. The zeros of $f'(x)$ are $x = -2$, 0.5 , and 3 .
 Point C has x co-ordinate of -0.9 and point B has x co-ordinate of 1.9 .

(i) For what range of values of x is $f(x)$ increasing?**1**(ii) Point C is the local maximum of $f'(x)$.**2**What type of point occurs on $y = f(x)$ when $x = -0.9$. Justify your answer.(iii) For what range of values of x is $f(x)$ concave down?**1****Question 9(c) continued on next page**

9(c)



The graphs of $y = \sin \frac{\pi}{2} x$ and $y = x^2$ are shown above.

The region bounded by these curves and the x -axis is shaded.

- (i) Show that the two curves meet where $x = 1$. 1
- (ii) Write the definite integral(s) which will give the volume of the solid of revolution when this region is rotated about the x -axis. **DO NOT EVALUATE THESE INTEGRALS** 1
- (iii) Use Simpson's Rule with five function values to approximate this volume, leaving your answer in terms of π . 3

Question 10 (12 Marks) Begin a SEPARATE sheet of paper**Marks**

- (a) The percentage of Carbon14 in an organism falls exponentially after the death of the organism. After 1845 years 80% of the original concentration of Carbon14 remains. Using the equation $C = C_0 e^{-kt}$ to represent the exponential fall of Carbon14:

- | | | |
|-------|---|----------|
| (i) | Find the exact value of k . | 2 |
| (ii) | An artefact containing this organism has 65% of the original concentration of Carbon14. How long has this organism been dead? Give answer to the nearest year. | 2 |
| (iii) | A sea sponge containing this organism has been dead for 12 000 years. What percentage (to 1 decimal place) of the original Carbon14 concentration does it have? | 2 |

- (b) Two sailors are paid to bring a motor launch back to Sydney from Gilligan's Island, a total distance of 1 200 km. They are each paid \$25 per hour for the time spent at sea.

The launch uses fuel at a rate of $20 + \frac{v^2}{10}$ litres per hour where the speed of the launch is v km

per hour. Diesel fuel costs \$1.25 per litre.

- | | | |
|-----|---|----------|
| (i) | Show that, to bring the launch back from Gilligan's Island, the total cost (\$ C) to the owners is | 3 |
|-----|---|----------|

$$C = \frac{90\,000}{v} + 150v.$$

- | | | |
|------|---|----------|
| (ii) | Find the speed that minimises the cost and determine this cost, giving your answer to the nearest dollar. | 3 |
|------|---|----------|

END OF EXAMINATION

SOLUTIONS TO 2 UNIT TRIAL MATHEMATICS

Question 1 (12 marks)

a) $\frac{e+1}{\pi} = 1.183565\dots$
 $\pi = 1.184$ (3 dp)

b) $\sin \theta = 0.500698846$

$\theta = 30^\circ 3'$

$\theta = 30^\circ$ (nearest degree)

c) Centre $(-2, 3)$

radius $= \sqrt{2.25} = 1.5$

d) $\frac{3+5+7+x}{4} = 6.75$

$15+x = 27$

$x = 12$

e) $|-3 - 4 \times 2.35| = |-12.4| = 12.4$

f) $3x^2 - 5x - 2 = (3x+1)(x-2)$

g) $2.32 \text{ rads} = 2.32 \times \frac{180^\circ}{\pi} = 132^\circ 56'$

h) $x^2 - 9 \geq 0$

D: $x \leq -3$ or $x \geq 3$

R: $y \geq 0$

Question 2 (12 marks)

a) (i) 0

(ii) $\frac{2x-3}{x^2-3x}$

(iii) $\frac{2 \cos x + (2x-1) \sin x}{\cos^2 x}$

b) $-2 \cos \frac{x}{2} + c$

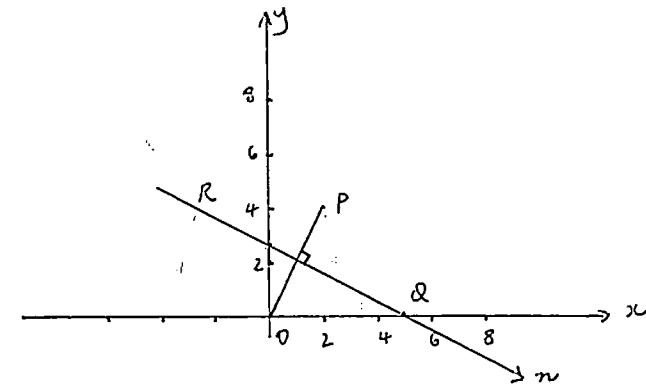
c) $\int_0^1 \frac{5}{6} e^{3x} dx = \frac{5}{6} \left[\frac{1}{3} e^{3x} \right]_0^1$
 $= \frac{5}{18} (e^3 - 1)$

d) (i) $\log pq = \log p + \log q = 1.75 + 2.25 = 4$

(ii) $\log \frac{q}{p} = \log q - \log p = 2.25 - 1.75 = 0.5$

(iii) Let $\sqrt[5]{pq^2} = y \Rightarrow \log_m \sqrt[5]{pq^2} = \log_m y$
 $\frac{1}{5} (\log p + 2 \log q) =$
 $\frac{1}{5} (1.75 + 2 \times 2.25) =$
 $\frac{1}{5} (1.75 + 4.5) = \log_m y$
 $\therefore m^{1.25} = y$

Question 3 (12 marks)



a) M is $(1, 2)$

b) Sub. $M(1, 2)$ into $x+2y=5$:

LHS $= 1 + 2 \times 2 = 5$

RHS $= 5$

$\therefore$ LHS = RHS

$\therefore M(1, 2)$ lies on $x+2y=5$

c) $m_{op} = \frac{4}{2} = 2$

d) $m_{line\ n} = -\frac{2 \frac{1}{2}}{5} = -\frac{1}{2}$

$m_{op} \times m_{line\ n} = 2 \times -\frac{1}{2} = -1$

$\therefore OP \perp$ line n

Also $(1, 2)$ is the midpoint of $OP \therefore$ bisector

$\therefore$ line n is the perpendicular bisector of OP .

e) $Q(5, 0)$

f) $m_{PQ} = -\frac{4}{3}$

$m_{OR} = m_{PQ} = -\frac{4}{3}$ ($PQ \parallel OR$)

By congruent triangles, and using diagram above,

R is $(-3, 4)$

g) $OQ \parallel PR$ (gradients 0)

$PQ \parallel OR$ (from f))

Diagonals meet at 90°

$\therefore PQOR$ is a rhombus (2 pairs of opposite sides parallel, diagonals meet at 90°)

NB: Other reasons are also possible.

(3)

Question 4 (12 marks)a) (i) Required to prove $(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2$

$$\begin{aligned} \text{LHS: } (m^2 - n^2)^2 + (2mn)^2 &= m^4 - 2m^2n^2 + n^4 + 4m^2n^2 \\ &= m^4 + 2m^2n^2 + n^4 \\ &= (m^2 + n^2)^2 \\ &= \text{RHS.} \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS}$$

(ii) If $m=3, n=2$: triad is 13, 5, 12b) (i) $f(x) = x - 2 \log_e x, x > 0$

$$f'(x) = 1 - \frac{2}{x} \text{ or } 1 - 2x^{-1}$$

$$f''(x) = 2x^{-2} \text{ or } \frac{2}{x^2}$$

(ii) For t.p. ($f'(x) = 0$): $1 - \frac{2}{x} = 0$

$$x - 2 = 0$$

$$x = 2$$

 $\therefore$ Possible t.p. is $(2, 2 - 2 \ln 2)$

To determine its nature use either 1st or 2nd derivative

Using 1st derivative:

x	1	2	3
$f'(x)$	-1	0	$\frac{1}{3}$

 $\therefore$ local minimum at $x=2$ Using 2nd derivative: at $x=2$

$$f''(2) = 2 \times 2^{-2} > 0$$

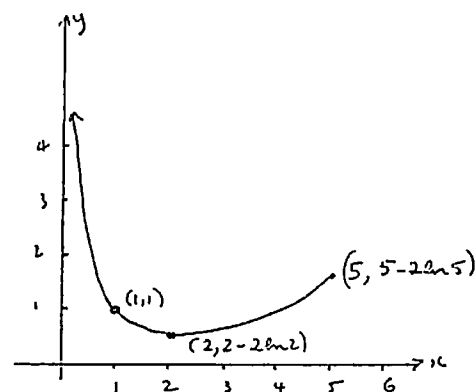
 $\therefore$ concave up $\therefore$ local minimum at $x=2$ (iii) For possible points of inflexion, $f''(x) = 0$

$$\therefore \frac{2}{x^2} = 0$$

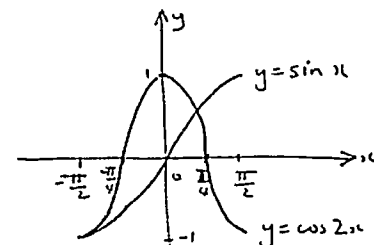
$$\text{but } \frac{2}{x^2} \neq 0$$

 $\therefore$ there are no points of inflexion.

(4)

(iv) at $x=1, f(1) = 1 - 2 \ln 1 = 1$ at $x=5, f(5) = 5 - 2 \ln 5 \approx 1.78$ $\therefore$ Maximum value is $5 - 2 \ln 5$ (v) $y = x - 2 \ln x, 0 < x \leq 5$ Question 5 (12 marks)

a) (i)

(ii) at $x = \frac{\pi}{6}, \sin \frac{\pi}{6} = \frac{1}{2}$

$$\cos 2 \times \frac{\pi}{6} = \cos \frac{\pi}{3} = \frac{1}{2}$$

 $\therefore \sin x = \cos 2x$ at $x = \frac{\pi}{6}$ (iii) $A = \int_{-\pi/2}^{\pi/6} (\cos 2x - \sin x) dx$

$$= \left[\frac{1}{2} \sin 2x + \cos x \right]_{-\pi/2}^{\pi/6}$$

$$= \left(\frac{1}{2} \sin \frac{\pi}{3} + \cos \frac{\pi}{6} \right) - \left(\frac{1}{2} \sin(-\pi) + \cos\left(-\frac{\pi}{2}\right) \right)$$

$$= \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{4} \text{ units}^2$$

(5)

Question 5 continued.

$$b) LHS = \frac{\sec \theta - \sec \theta \cos^4 \theta}{1 + \cos^2 \theta}$$

$$= \frac{\sec \theta (1 - \cos^4 \theta)}{1 + \cos^2 \theta}$$

$$= \frac{\sec \theta (1 + \cos^2 \theta)(1 - \cos^2 \theta)}{1 + \cos^2 \theta}$$

$$= \frac{1}{\cos \theta} \cdot \sin^2 \theta$$

$$= \frac{\sin \theta}{\cos \theta} \cdot \sin \theta$$

$$= \tan \theta \cdot \sin \theta$$

$$= RHS$$

$$c) (i) \Delta = (2-k)^2 - 4 \cdot 1 \cdot (2 \cdot 25)$$

$$= (2-k)^2 - 9$$

For equal roots, $\Delta = 0$

$$\therefore (2-k)^2 - 9 = 0$$

$$(2-k)^2 = 9$$

$$2-k = \pm 3$$

$$k = 5 \text{ or } -1$$

$$(ii) y = kx + 1$$

$$y = x^2 + 2x + 3 \cdot 25$$

$$\text{Solve: } x^2 + 2x + 3 \cdot 25 = kx + 1$$

$$x^2 + (2-k)x + 2 \cdot 25 = 0$$

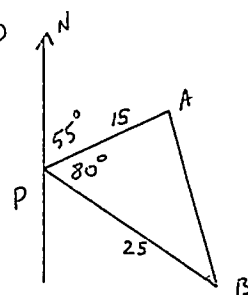
For line to be a tangent, $\Delta = 0$, then from (i)

$$k = 5 \text{ or } -1.$$

(6)

Question 6 (12 marks)

a) (i)



$$(ii) AB^2 = 15^2 + 25^2 - 2 \cdot 15 \cdot 25 \cdot \cos 80^\circ$$

$$= 719.76$$

$$AB = 26.8 \text{ km}$$

$$b) \sum_{r=3}^7 2^r - 3r = 2^3 - 3 \times 3 + 2^4 - 3 \times 4 + 2^5 - 3 \times 5$$

$$+ 2^6 - 3 \times 6 + 2^7 - 3 \times 7$$

$$= (2^3 + 2^4 + 2^5 + 2^6 + 2^7) - 3(3 + 4 + 5 + 6 + 7)$$

$$= 248 - 75 = 173$$

$$c) (i) \text{ When } t=0, R = 6(e^0 + e^0) = 12 \text{ L/min}$$

$$(ii) V = \int 6(e^t + e^{-t}) dt$$

$$V = 6(e^t - e^{-t}) + c$$

$$\text{When } t=0, V=0 \Rightarrow 0 = 6(e^0 - e^0) + c$$

$$c = 0$$

$$V = 6(e^t - e^{-t})$$

$$(iii) 5 = 6(e^t - \frac{1}{e^t})$$

$$5e^t = 6e^{2t} - 6$$

$$\therefore 6e^{2t} - 5e^t - 6 = 0$$

$$(iv) (2e^t - 3)(3e^t + 2) = 0$$

$$e^t = \frac{3}{2} \text{ or } e^t = -\frac{2}{3}$$

$$\text{but } e^t \neq -\frac{2}{3} \therefore \text{no soln.}$$

$$\therefore t = \ln \frac{3}{2} \div 24 \text{ seconds}$$

for diagram
for $\angle APB$
 $= 80^\circ$ -1/2 if not
to 1 d.p.NB:
It is not
intended
that
rules for
sum of AP
& GP's
be used
here.-1/2 if no
units are
provided.-1/2 if not
to nearest
second.

(7)

Question 7 (12 marks)

$$a) \frac{dy}{dx} = \frac{3 \sec^2 x}{\tan x}$$

$$y = 3 \ln(\tan x) + c$$

$$\text{When } x = \frac{\pi}{4}, y = 4 \Rightarrow 4 = 3 \ln(\tan \frac{\pi}{4}) + c$$

$$\therefore c = 4$$

$$\therefore y = 3 \ln(\tan x) + 4 \quad [2]$$

$$b) (i) 1 \text{ year} = 365 \times 24 \times 60 \text{ minutes} \\ = 525600 \text{ minutes}$$

$$\therefore \text{Incremental rate of interest} = \frac{12}{525600} \% \text{ per minute}$$

$$= \frac{0.12}{525600} \text{ per minute}$$

$$\div 0.00002283 \text{ per min.} \quad \frac{1}{2}$$

Since interest is added onto the amount invested then interest is multiplied by approximately

$$1.00002283 \text{ every minute} \quad [2] \quad \frac{1}{2}$$

$$(ii) 1000 (1.00002283)^{525600} \\ = \underline{\$1127} \text{ (nearest dollar)} \quad [1]$$

$$c) (i) \angle QOP = \frac{2\pi}{3} \quad (\text{angle sum about a point is } 360^\circ) \\ \angle QPO = \frac{2\pi - \frac{2\pi}{3}}{2} = \frac{\pi}{6} \quad (\text{angle sum of triangle is } 360^\circ; \text{ angles opposite equal sides are equal}) \quad [2]$$

$$(ii) A_{\text{sector OPR}} = \frac{1}{2} \times 5\sqrt{3} \times 5\sqrt{3} \times \frac{\pi}{3} = \frac{25\pi}{2} \text{ unit}^2 \quad [1]$$

$$(iii) A_{\text{sector OQR}} = \frac{1}{2} \times 5 \times 5 \times \frac{2\pi}{3} = \frac{25\pi}{3} \text{ units}^2 \quad [1]$$

(iv) There are several ways to do this which may involve areas of segments/triangles/sectors.

PTO.

(8)

Question 7 (continued)

Method 1

$$(iv) A_{\text{segment } QC_1R} = \frac{1}{2} \times (5\sqrt{3})^2 \left(\frac{\pi}{3} - \sin \frac{\pi}{3} \right) \\ = 25 \times \frac{3}{2} \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

$$= 25 \left(\frac{\pi}{2} - \frac{3\sqrt{3}}{4} \right)$$

$$A_{\text{segment } QC_2R} = \frac{1}{2} \times 5^2 \left(\frac{2\pi}{3} - \sin \frac{2\pi}{3} \right)$$

$$= 25 \times \frac{1}{2} \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

$$= 25 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right)$$

$$A_{\text{shaded}} = 25 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right) - 25 \left(\frac{\pi}{2} - \frac{3\sqrt{3}}{4} \right)$$

$$= 25 \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6} \right) \text{ units}^2 \quad [3]$$

Alt. Method 2

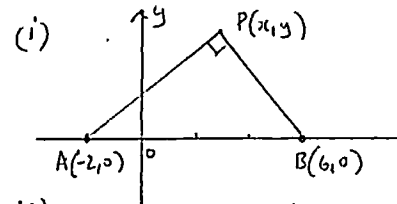
$$\text{Shaded area} = (iii) - (ii) + 2 \Delta's$$

$$= \frac{25\pi}{3} - \frac{25\pi}{2} + 2 \times \frac{25\sqrt{3}}{4}$$

$$= 25 \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6} \right) \quad (13)$$

Question 8 (12 marks)

a)



$$m_{PA} = \frac{y-0}{x-(-2)} = \frac{y}{x+2}$$

[1]

$$(ii) m_{PA} \times m_{PB} = -1 \\ \frac{y}{x+2} \times \frac{y}{x-6} = -1 \\ y^2 = -(x+2)(x-6) \\ = -x^2 + 4x + 12$$

$$x^2 - 4x + 4 + y^2 = 12 + 4$$

$$(x-2)^2 + y^2 = 16$$

$\therefore$ Locus is a circle, centre (2, 0), radius 4 units. [2]

(9)

Question 8 (continued)

$$b) V = 6t - 8 - t^2 = \frac{dx}{dt}$$

$$(i) x = 3t^2 - 8t - \frac{t^3}{3} + c$$

$$\text{When } t=0, x = +5 \Rightarrow \underline{5 = c}$$

$$\therefore x = 3t^2 - 8t - \frac{t^3}{3} + 5$$

$$(ii) \text{ At rest, } v=0 \Rightarrow 0 = 6t - 8 - t^2$$

$$0 = (t-4)(t-2)$$

$$\therefore \text{ at rest when } t=2 \text{ and } t=4$$

$$(iii) x(0) = 5$$

$$x(2) = -1\frac{2}{3}$$

$$\therefore \text{ In first 2 secs it travels } 5 + 1\frac{2}{3} = 6\frac{2}{3} \text{ m}$$

$$x(4) = x = -\frac{1}{3}$$

$$\therefore \text{ from } t=2 \text{ to } t=4 \text{ it travels } 1\frac{1}{3} \text{ units.}$$

$$\therefore \text{ distance travelled} = 5 + 1\frac{2}{3} + 1\frac{1}{3} = \underline{8 \text{ units}}$$

$$c) (i) P(6,6) = \frac{3}{8} \times \frac{3}{8} = \frac{9}{64}$$

$$(ii) P(\tilde{6}, \tilde{6}) = \frac{5}{8} \times \frac{5}{8} = \frac{25}{64}$$

$$(iii) P(6, \tilde{6}) + P(\tilde{6}, 6) = \frac{3}{8} \times \frac{5}{8} \times 2 = \frac{30}{64} = \frac{15}{32}$$

Question 9 (12 marks)

$$a) (i) x^2 - 6x + 8 = 2y$$

$$x^2 - 6x + 9 - 1 = 2y$$

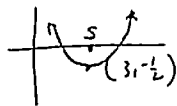
$$(x-3)^2 = 2y + 1$$

$$= 2(y + \frac{1}{2})$$

$$(x-3)^2 = 4(\frac{1}{2})(y - (-\frac{1}{2}))$$

$$(ii) \text{ Vertex} = (3, -\frac{1}{2}) \quad a = \frac{1}{2}$$

$$\text{Focus} = (3, 0)$$



(10)

Question 9 (continued)

$$b) (i) \text{ For increasing function, } f'(x) > 0$$

$$\text{ie } \{x: -2 < x < \frac{1}{2}\}, \{x: x > 3\}$$

$$(ii) f'(-0.9) \text{ is a maximum, so } f''(-0.9) = 0$$

$$\therefore C \text{ is a possible point of inflexion.}$$

$$\text{for } x < -0.9, f'(x) > 0$$

$$\text{for } x > -0.9, f'(x) > 0$$

$$\therefore \text{ In the neighbourhood of } -0.9, \text{ gradients } > 0.$$

$$\therefore C \text{ is a point of inflexion.}$$

$$(iii) \text{ For concave down, } f'(x) \text{ is decreasing.}$$

$$\therefore \{x: -0.9 < x < 1.9\}$$

$$c) (i) \text{ at } x=1, \sin \frac{\pi}{2} \times 1 = \sin \frac{\pi}{2} = 1$$

$$1^2 = 1$$

$$\therefore \sin \frac{\pi}{2} x = x^2 \text{ at } x=1$$

$$(ii) V = \pi \int_0^1 (x^2)^2 dx - \pi \int_1^2 \left(\sin \frac{\pi}{2} x\right)^2 dx$$

OR

$$V = \pi \int_0^1 x^4 dx - \pi \int_1^2 \sin^2 \frac{\pi}{2} x dx$$

x	Weight	$f(x)^2$
0	1	0
$\frac{1}{2}$	4	$\frac{1}{16}$
1	2	1
$1\frac{1}{2}$	4	$\sin^2 \frac{3\pi}{4} = \frac{1}{2}$
2	1	$\sin^2 \pi = 0$

$$V = \pi \times \frac{2-0}{12} \left[0 + 0 + 4\left(\frac{1}{16} + \frac{1}{2}\right) + 2(1) \right]$$

$$= \frac{\pi}{6} \left[\frac{17}{4} \right]$$

$$= \frac{17\pi}{24} \text{ u}^3$$

-1/2 if either is missing

(11)

Question 10 (12 Marks)

a) (i) When $t=0$, $A=A_0$ When $t=1845$, $A=0.8(A_0)$

$$\therefore 0.8 A_0 = A_0 e^{-1845k}$$

$$0.8 = e^{-1845k}$$

$$\ln 0.8 = -1845k$$

$$k = -\frac{\ln 0.8}{1845} \text{ or } \frac{\ln 1.25}{1845}$$

[2]

(ii) $A = 0.65 A_0$ when $t=T$

$$\therefore 0.65 A_0 = A_0 e^{+\frac{\ln 0.8}{1845} T}$$

$$\ln 0.65 = \frac{\ln 0.8}{1845} T$$

$$T = \frac{1845 \times \ln 0.65}{\ln 0.8}$$

$$T = 3562 \text{ years}$$

[2]

(iii) When $t=12000$:

$$A = A_0 e^{+12000 \times \frac{\ln 0.8}{1845}}$$

$$A \doteq 0.234 A_0$$

$\therefore A \doteq 23.4\%$ of original amount is left

[2]

b) $S = \frac{D}{T}$ (i) $T = \frac{D}{\frac{r}{100}} = \frac{1200}{r}$ hours

$$\text{Cost of wages} = 25 \times 2 \times \frac{1200}{r} = \frac{60000}{r}$$

$$\text{Cost of fuel} = \left(20 + \frac{r^2}{10}\right) \times \frac{1200}{r} \times 1.25$$

$$= \left(20 + \frac{r^2}{10}\right) \times \frac{1500}{r}$$

$$= \frac{30000}{r} + 150r$$

$$\text{Total cost, } C = \frac{60000}{r} + \frac{30000}{r} + 150r$$

$$C = \frac{90000}{r} + 150r$$

[3]

(12)

Question 10 (continued)

$$(ii) C = \frac{90000}{r} + 150r$$

$$\frac{dC}{dr} = 150 - 90000r^{-2}$$

For min. cost, $\left(\frac{dC}{dr} = 0\right)$:

$$150 - \frac{90000}{r^2} = 0$$

$$150r^2 - 90000 = 0$$

$$r^2 - 600 = 0$$

$$r = \sqrt{600} \quad (r > 0)$$

$$\therefore r = 10\sqrt{6}$$

check $C(10\sqrt{6})$ is a minimum:

$$\frac{d^2C}{dr^2} = 180000r^{-3}$$

$$\text{at } r = 10\sqrt{6}, \quad \frac{d^2C}{dr^2} = \frac{180000}{(10\sqrt{6})^3} > 0$$

 $\Rightarrow$ concave up $\therefore$ local min at $r = 10\sqrt{6}$.

(May use 1st derivative to test nature of stationary point.)

$$\therefore \text{Minimum cost} = \frac{90000}{\sqrt{600}} + 150 \times \sqrt{600}$$

$$= \$7348$$

[3]

END OF SOLUTIONS